

EXCEL

1. Create statistical summary on relevant features such as number of labs procedures, number of procedures, and number of medications

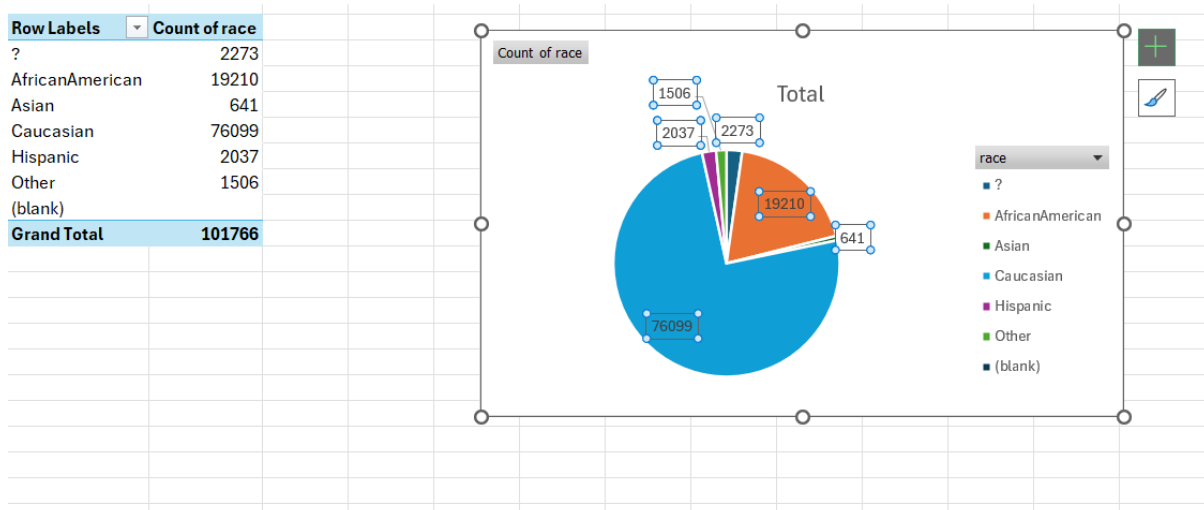
Column1	
Mean	12522
Standard E	0
Median	12522
Mode	#N/A
Standard D	#DIV/0!
Sample Va	#DIV/0!
Kurtosis	#DIV/0!
Skewness	#DIV/0!
Range	0
Minimum	12522
Maximum	12522
Sum	12522
Count	1

2. Create a report showing the number of readmissions by gender and age group

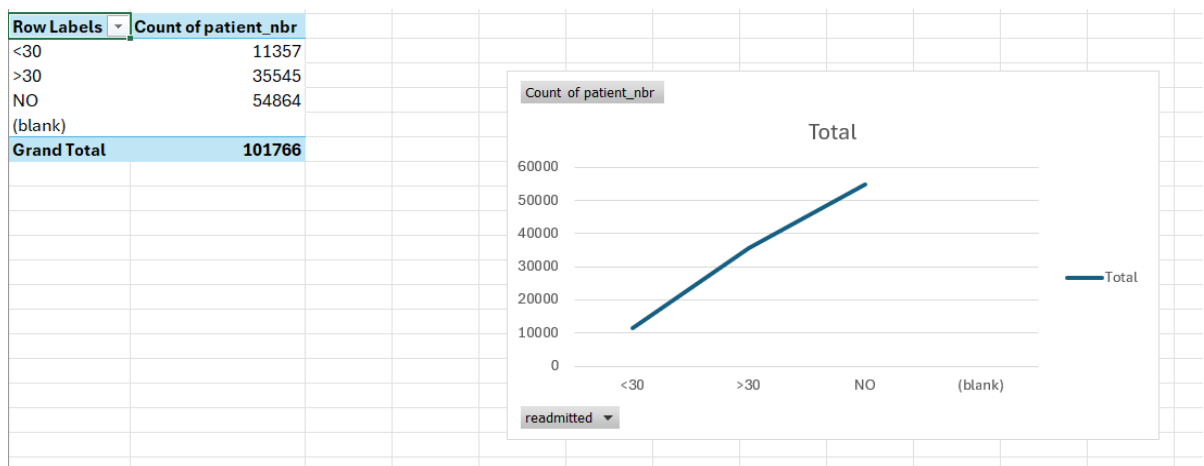
Count of encounter_id	Column Labels				
Row Labels	<30	>30	NO	(blank)	Grand Total
Female	6152	19518	29038		54708
Male	5205	16027	25823		47055
Unknown/Invalid (blank)			3		3
Grand Total	11357	35545	54864		101766

Count of encounter_id	Column Labels			
Row Labels	<30	>30	NO	(blank) Grand Total
[0-10)	3	26	132	161
[10-20)	40	224	427	691
[20-30)	236	510	911	1657
[30-40)	424	1187	2164	3775
[40-50)	1027	3278	5380	9685
[50-60)	1668	5917	9671	17256
[60-70)	2502	7897	12084	22483
[70-80)	3069	9475	13524	26068
[80-90)	2078	6223	8896	17197
[90-100)	310	808	1675	2793
(blank)				
Grand Total	11357	35545	54864	101766

3. Race Distribution in a chart

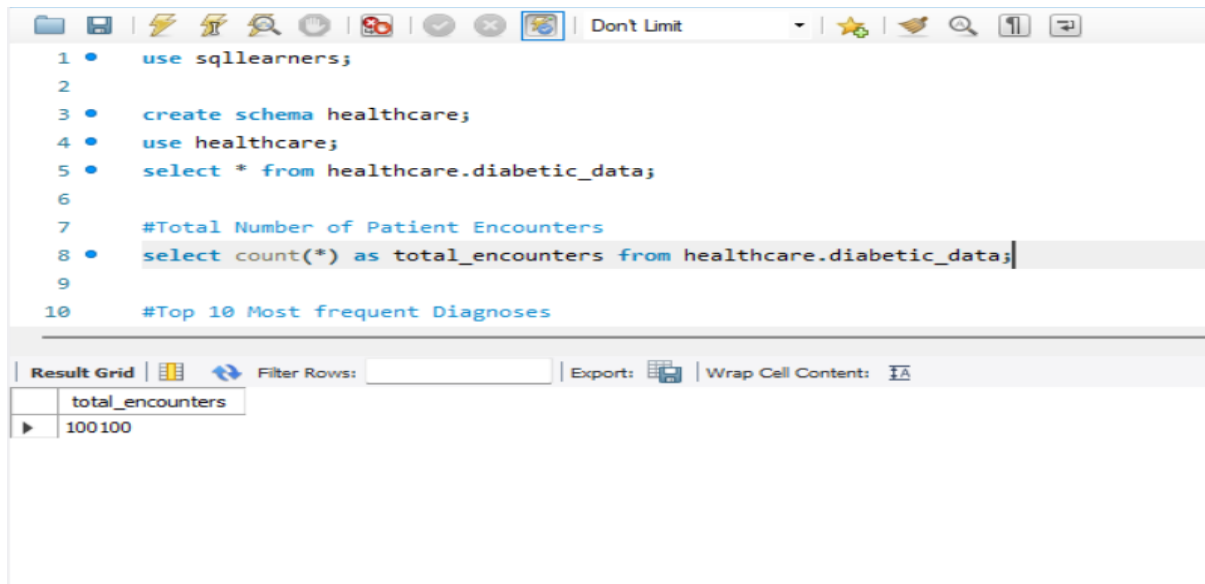


4. Plot count of patient for each category of readmitted



SQL

Calculate the total number of patient encounters in the healthcare dataset



```

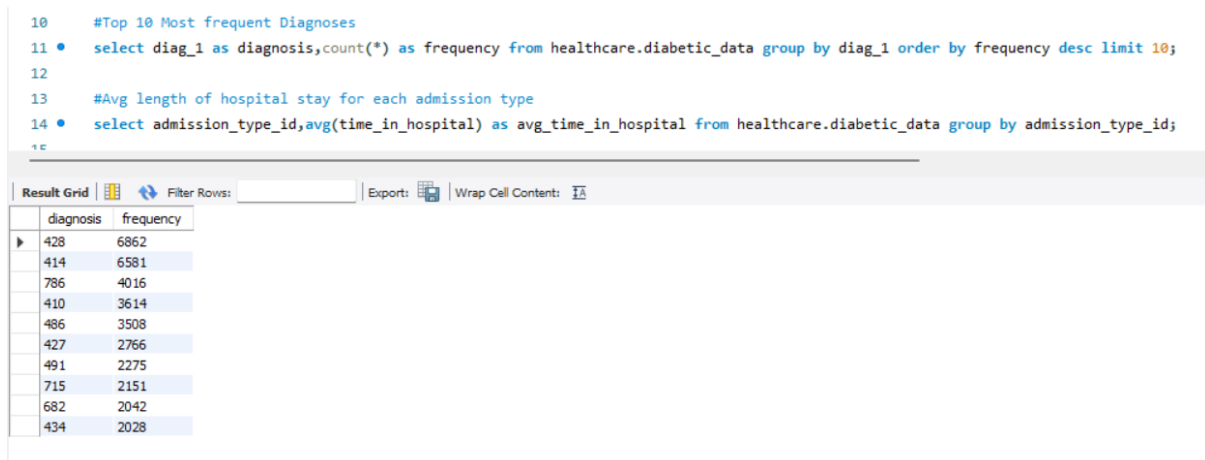
1 • use sqllearners;
2
3 • create schema healthcare;
4 • use healthcare;
5 • select * from healthcare.diabetic_data;
6
7 #Total Number of Patient Encounters
8 • select count(*) as total_encounters from healthcare.diabetic_data;
9
10 #Top 10 Most frequent Diagnoses

```

Result Grid

total_encounters
100100

Identify the top 10 most frequent diagnoses in the dataset



```

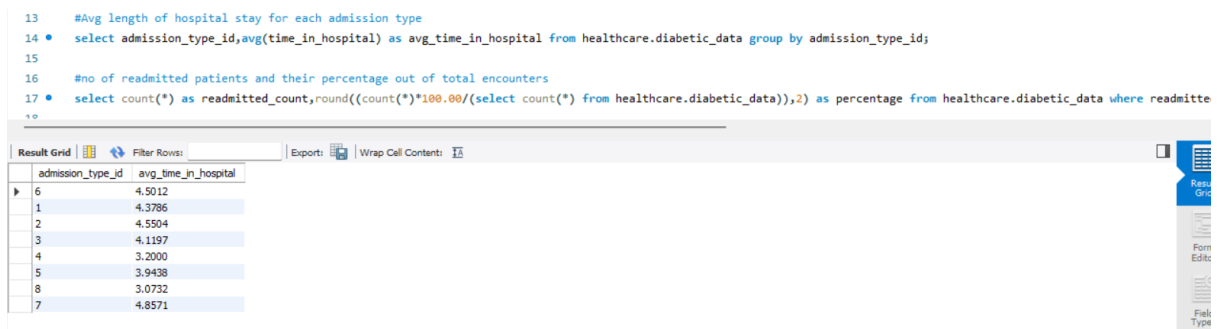
10 #Top 10 Most frequent Diagnoses
11 • select diag_1 as diagnosis,count(*) as frequency from healthcare.diabetic_data group by diag_1 order by frequency desc limit 10;
12
13 #Avg length of hospital stay for each admission type
14 • select admission_type_id,avg(time_in_hospital) as avg_time_in_hospital from healthcare.diabetic_data group by admission_type_id;
15
16 #no of readmitted patients and their percentage out of total encounters
17 • select count(*) as readmitted_count,round((count(*)*100.00/(select count(*) from healthcare.diabetic_data)),2) as percentage from healthcare.diabetic_data where readmitte

```

Result Grid

diagnosis	frequency
428	6862
414	6581
786	4016
410	3614
486	3508
427	2766
491	2275
715	2151
682	2042
434	2028

Calculate the average length of hospital stay for each admission type



```

13 #Avg length of hospital stay for each admission type
14 • select admission_type_id,avg(time_in_hospital) as avg_time_in_hospital from healthcare.diabetic_data group by admission_type_id;
15
16 #no of readmitted patients and their percentage out of total encounters
17 • select count(*) as readmitted_count,round((count(*)*100.00/(select count(*) from healthcare.diabetic_data)),2) as percentage from healthcare.diabetic_data where readmitte

```

Result Grid

admission_type_id	avg_time_in_hospital
6	4.5012
1	4.3786
2	4.5504
3	4.1197
4	3.2000
5	3.9438
8	3.0732
7	4.8571

Determine the number of readmitted patients and the percentage of total encounters that they represent

```

16  #no of readmitted patients and their percentage out of total encounters
17  • select count(*) as readmitted_count,round((count(*)*100.00/(select count(*) from healthcare.diabetic_data)),2) as percentage from healthcare.diabetic_data where readmitted = 'NO';
18
19  #age distribution of patients

```

readmitted_count	percentage
54034	53.98

identify the most common procedures performed during patient encounters

```

23  #most common procedures
24  • select medical_specialty,count(*) as procedure_count from healthcare.diabetic_data where medical_specialty is not null group by medical_specialty order by procedure_count desc limit 5;

```

medical_specialty	procedure_count
?	49240
InternalMedicine	14422
Emergency/Trauma	7561
Family/GeneralPractice	7388
Cardiology	5311

Calculate the average number of medications prescribed for patients in each age group

```

25  #avg no of medications prescribes for patients accordin
26  • select age,avg(num_medications) as avg_no_of_medications from healthcare.diabetic_data group by age order by age;

```

age	avg_no_of_medications
[0-10)	6.1125
[10-20)	8.2710
[20-30)	11.9485
[30-40)	14.0818
[40-50)	15.3835
[50-60)	16.5804
[60-70)	17.1794
[70-80)	16.4307
[80-90)	15.3503
[90-100)	13.8231

Identify the age distribution of patients

```

19  #age distribution of patients
20  • select count(*) as patient_count,age from healthcare.diabetic_data group by age order by age;
21

```

patient_count	age
160	[0-10)
690	[10-20)
1650	[20-30)
3753	[30-40)
9588	[40-50)
17018	[50-60)
22083	[60-70)
25528	[70-80)
16872	[80-90)
2758	[90-100)

Identify the distribution of readmission rates across different payer codes

```
28 #readmission rates across different payer codes
29 • SELECT payer_code,sum(CASE WHEN readmitted != 'NO' THEN 1 ELSE 0 END) AS readmitted_count,count(*) AS total_encounters,
30 round((SUM(CASE WHEN readmitted != 'NO' THEN 1 ELSE 0 END) * 100.0) / COUNT(*), 2) AS readmission_rate
31 FROM healthcare.diabetic_data
32 GROUP BY payer_code
33 ORDER BY readmission_rate DESC;
34
```

Result Grid			
Filter Rows:			
Export:			
Wrap Cell Content:			
payer_code	readmitted_count	total_encounters	readmission_rate
MP	48	79	60.76
DM	282	546	51.65
MC	15546	31955	48.65
SP	2358	4968	47.46
MD	1665	3509	47.45
OT	45	95	47.37
HM	2937	6207	47.32
?	18003	39368	45.73
OG	465	1026	45.32
CM	853	1914	44.57
SI	21	51	41.18
CP	960	2480	38.71
UN	914	2427	37.66
BC	1710	4618	37.03
CH	46	145	31.72
PO	184	580	31.72
WC	29	131	22.14
FR	0	1	0.00

PYTHON

Perform descriptive statistical analysis for numerical features

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
```

```
df = pd.read_csv("diabetic_data.csv")
```

```
import numpy as np
import pandas as pd
import scipy.stats as sp
```

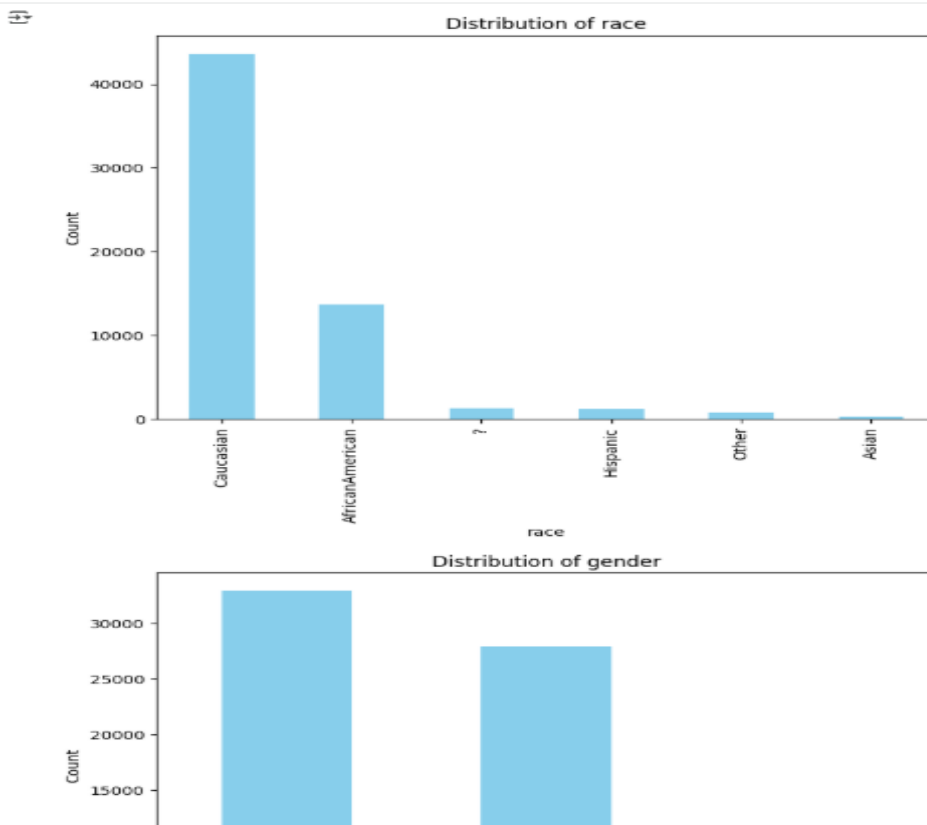
```
#task 1
numerical_stats = df.select_dtypes([float,int]).describe()
print("Descriptive Statistics for Numerical Features:")
numerical_stats
```

Descriptive Statistics for Numerical Features:

	encounter_id	patient_nbr	admission_type_id	discharge_disposition_id	admission_source_id	time_in_hospital	num_lab_procedures	num_procedures	num_medications	number_outpatient	number_emergency	number_inpatient
count	6.078800e+04	6.078800e+04	60788.000000	60788.000000	60788.000000	60788.000000	60788.000000	60788.000000	60788.000000	60788.000000	60788.000000	60788.000000
mean	9.671185e+07	4.146263e+07	2.175380	4.131391	6.030911	4.528558	42.749655	1.348983	15.509360	0.256778	0.139189	0.595956
std	4.709115e+07	3.559434e+07	1.572949	5.850999	4.588237	3.062870	19.083978	1.676017	8.170832	0.947908	0.646483	1.213157
min	1.252200e+04	1.350000e+02	1.000000	1.000000	1.000000	1.000000	1.000000	0.000000	1.000000	0.000000	0.000000	0.000000
25%	5.851407e+07	9.393350e+06	1.000000	1.000000	1.000000	2.000000	31.000000	0.000000	10.000000	0.000000	0.000000	0.000000
50%	9.831163e+07	2.492085e+07	2.000000	1.000000	7.000000	4.000000	44.000000	1.000000	14.000000	0.000000	0.000000	0.000000
75%	1.408804e+08	7.449455e+07	3.000000	5.000000	7.000000	6.000000	56.000000	2.000000	19.000000	0.000000	0.000000	1.000000
max	1.701039e+08	1.152185e+08	8.000000	28.000000	20.000000	14.000000	129.000000	6.000000	81.000000	36.000000	42.000000	21.000000

Visualize the distribution of categorical features - race and gender

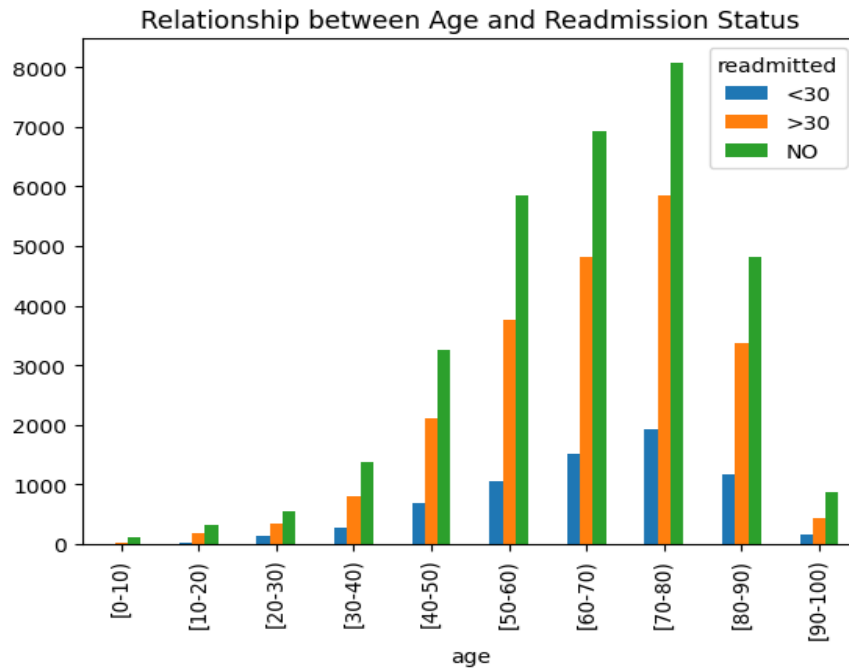
```
#task 2
categorical_features = ['race', 'gender']
for feature in categorical_features:
    plt.figure(figsize=(8, 6))
    df[feature].value_counts().plot(kind='bar', color='skyblue')
    plt.title(f'Distribution of {feature}')
    plt.xlabel(feature)
    plt.ylabel('Count')
    plt.show()
```



Explore the relationship between readmission status and age

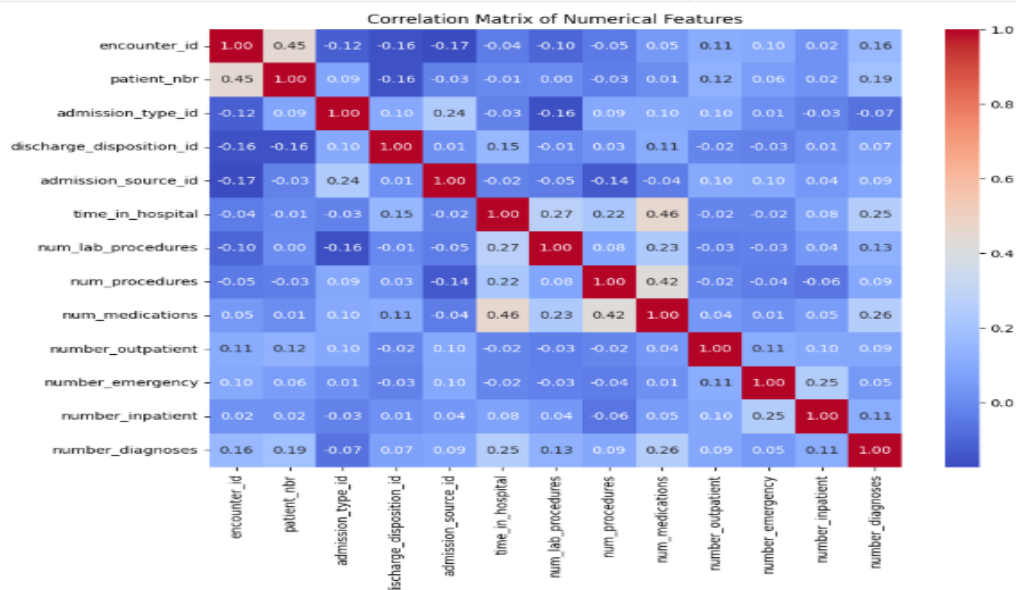
```
#task 3
counts = pd.crosstab(df.age,df.readmitted)
plt.figure(figsize=(10, 6))
counts.plot(kind='bar')
plt.title('Relationship between Age and Readmission Status')
plt.show()
```

<Figure size 1000x600 with 0 Axes>



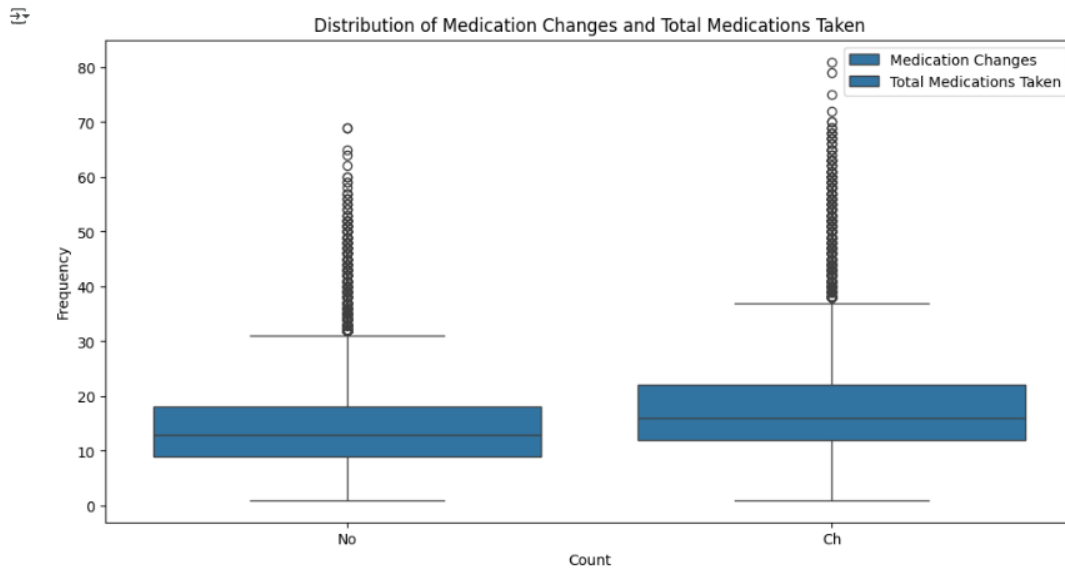
Investigate correlations between numerical features

```
#task 4
correlation_matrix = df.select_dtypes([float,int]).corr()
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", annot_kws={"size": 10})
plt.title('Correlation Matrix of Numerical Features')
plt.show()
```



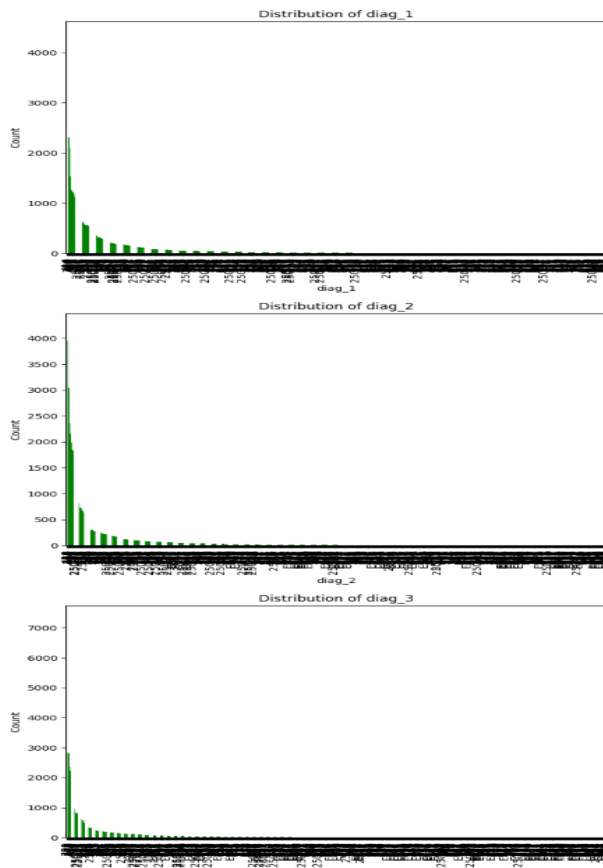
Analyse the distribution of medication changes and total medications taken

```
#task 5
plt.figure(figsize=(12, 6))
sns.boxplot(data=df, x='change', y='num_medications')
plt.title('Distribution of Medication Changes and Total Medications Taken')
plt.xlabel('Count')
plt.ylabel('Frequency')
plt.legend(['Medication Changes', 'Total Medications Taken'])
plt.show()
```



Examine the distribution of diagnoses categories

```
#task 6
diagnosis_features = ['diag_1', 'diag_2', 'diag_3']
for feature in diagnosis_features:
    plt.figure(figsize=(8, 6))
    df[feature].value_counts().plot(kind='bar', color='green')
    plt.title(f'Distribution of {feature}')
    plt.xlabel(feature)
    plt.ylabel('Count')
    plt.show()
```

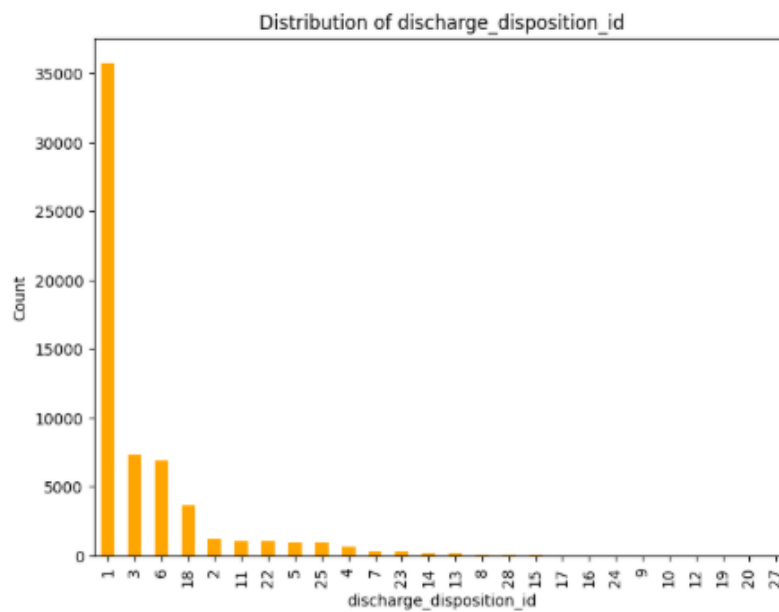
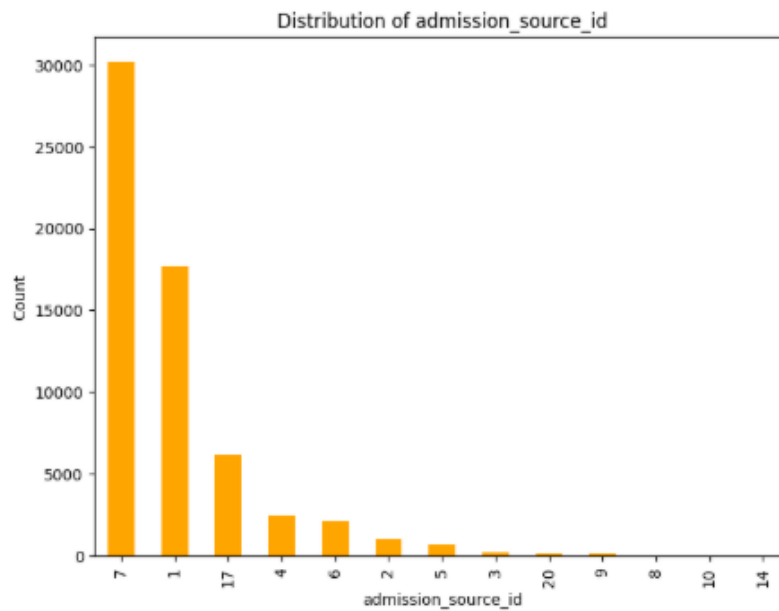
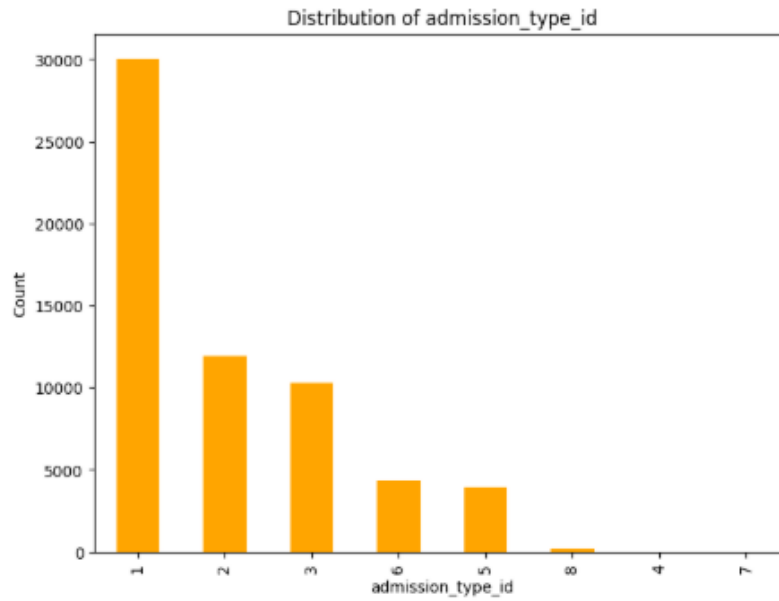



Explore the distribution of patients across admission types, sources, and discharge dispositions



#task 7

```
admission_features = ['admission_type_id', 'admission_source_id', 'discharge_disposition_id']
for feature in admission_features:
    plt.figure(figsize=(8, 6))
    df[feature].value_counts().plot(kind='bar', color='orange')
    plt.title(f'Distribution of {feature}')
    plt.xlabel(feature)
    plt.ylabel('Count')
    plt.show()
```

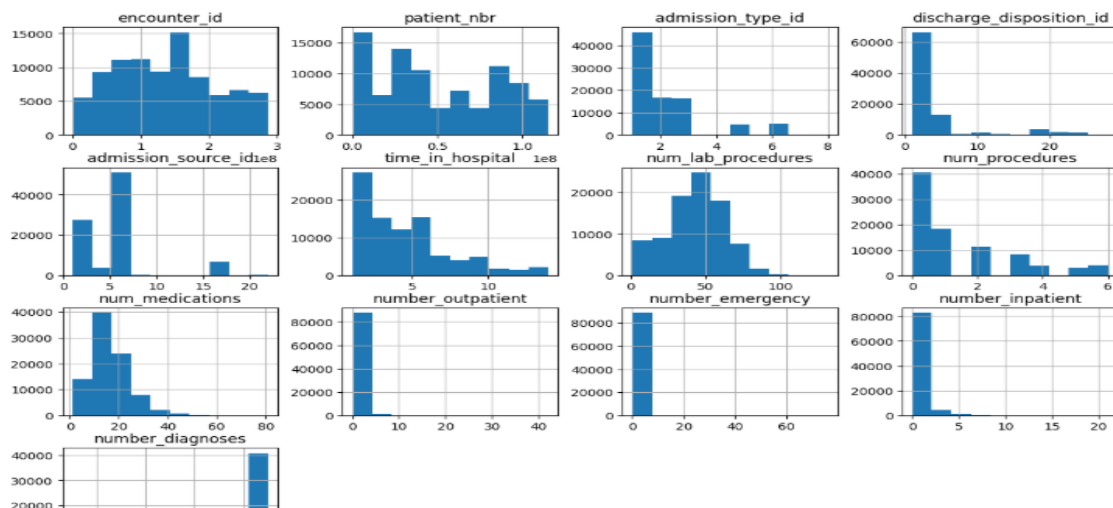
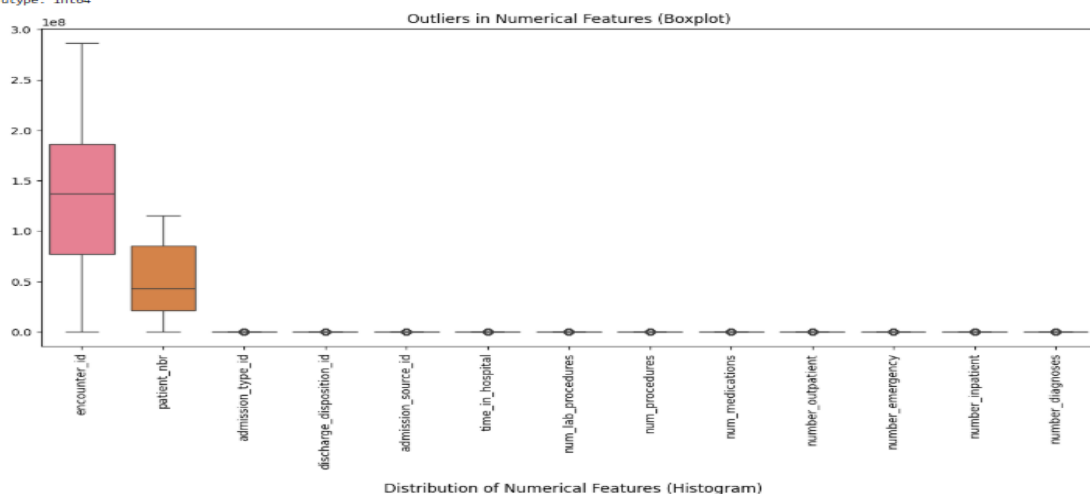


Identify and visualize any outliers in the dataset, especially in numerical features



```
numeric_df = df.select_dtypes(include=["int64", "float64"])
print("Numerical columns selected:\n", numeric_df.columns)
Q1 = numeric_df.quantile(0.25)  # 25th percentile
Q3 = numeric_df.quantile(0.75)  # 75th percentile
IQR = Q3 - Q1
outliers = (numeric_df < (Q1 - 1.5 * IQR)) | (numeric_df > (Q3 + 1.5 * IQR))
print("\nOutlier count per numerical feature:")
print(outliers.sum())
plt.figure(figsize=(14, 6))
sns.boxplot(data=numeric_df)
plt.title("Outliers in Numerical Features (Boxplot)")
plt.xticks(rotation=90)
plt.show()
numeric_df.hist(figsize=(14, 10))
plt.suptitle("Distribution of Numerical Features (Histogram)")
plt.show()
```

```
num_procedures      4896
num_medications      2268
number_outpatient    14861
number_emergency     9346
number_inpatient     6165
number_diagnoses      289
dtype: int64
```



Exploratory Data Analysis (EDA) Report – Healthcare Fraud Detection System

1. Introduction

Healthcare fraud leads to unnecessary medical costs and misuse of insurance claims. Before building a fraud detection model, **Exploratory Data Analysis (EDA)** is performed to understand the data, reveal patterns, and detect anomalies that may indicate fraudulent activity.

2. Objective of EDA

- Understand data structure and identify missing or incorrect values.
 - Detect patterns related to suspicious claims.
 - Identify outliers and anomalies (e.g., unusually high number of tests or procedures).
 - Support feature selection for the fraud detection model.
-

3. Tools Used

Python libraries:

- pandas for data loading and cleaning
 - matplotlib / seaborn for visualization
 - numpy for numerical analysis
-

4. Steps Performed

a) Data Loading & Overview

```
df = pd.read_csv("healthcare_fraud.csv")
```

```
df.info()
```

```
df.describe()
```

Reviewed column types, missing values, and summary statistics.

b) Data Cleaning

- Removed duplicates
- Filled missing numeric values with median; categorical with "Unknown".

c) Outlier Detection

Used the IQR (Interquartile Range) method to find unusually high lab tests, procedures, or visits.

```
sns.boxplot(df["num_lab_procedures"])
```

Outliers may indicate excessive billing or unnecessary procedures.

d) Visualization & Insights

- Histograms showed skewness in number of procedures and medications.
 - Boxplots revealed extreme values (potential fraud cases).
 - Correlation heatmap showed strong relationships between inpatient visits, emergency visits, and lab procedures.
-

5. Key Insights

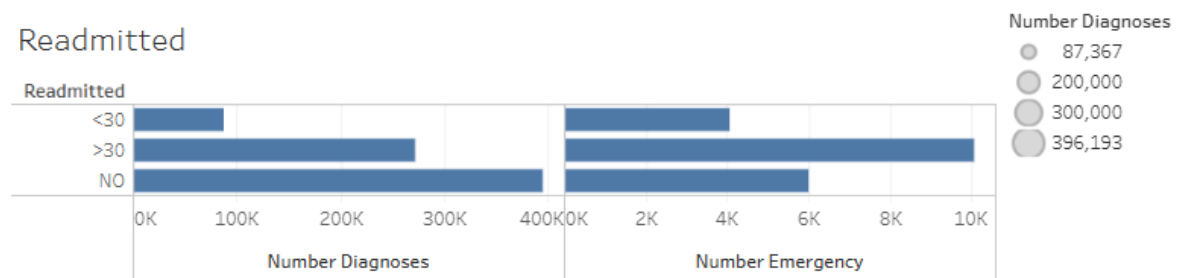
Finding	Interpretation
Very high lab procedures and multiple emergency visits for some claims	Possible overbilling or staged fraud
Strong correlation between inpatient, emergency, and outpatient visits	May indicate excessive hospital usage for claim inflation
High medication intensity (medications per day)	May indicate prescription misuse

6. Conclusion

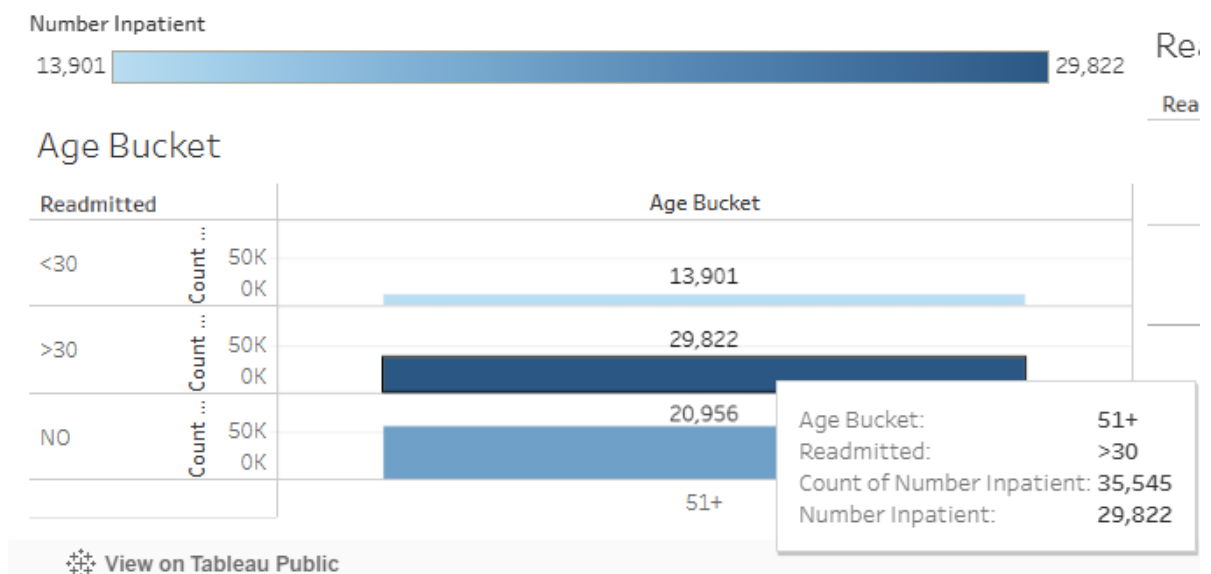
EDA helped identify suspicious claim patterns and outliers that are likely indicators of fraud. These insights will guide feature selection and improve machine-learning-based fraud detection accuracy.

Tableau

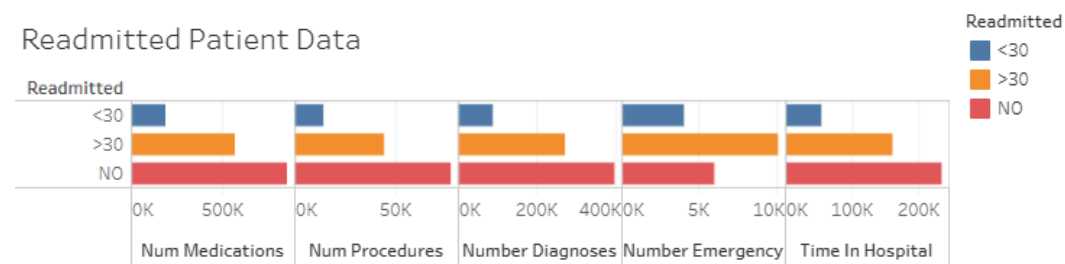
Show the distribution of readmitted counts of diagnoses and emergency cases through bar graphs



Create a visual capturing age bucket count by readmitted

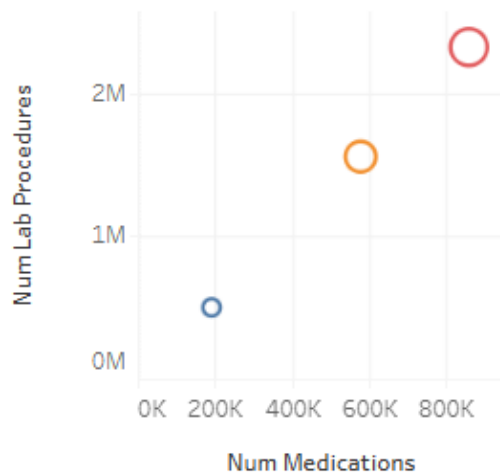


Visualize the number of medications, procedures, diagnoses, emergencies, time in the hospital by readmitted

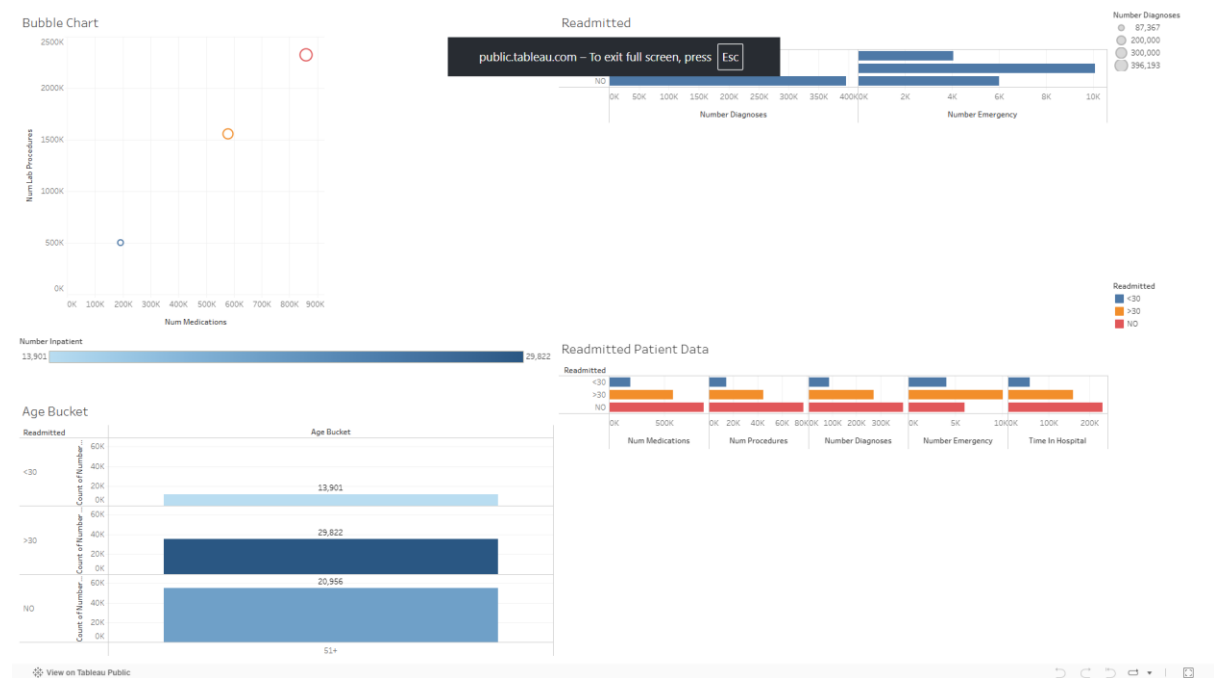


Create a bubble chart showing the number of medications, Lab procedures and Diagnoses and use Readmitted as color code

Bubble Chart



DASHBOARD



Hi There

Check out my project at tableau Public : https://public.tableau.com/views/IITK-CapstoneProject/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Thank You For Your Guidance
Sriram Kancharla